

Assignment 2: Stacks & Queues

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Part 1

Testing ArrayStack:

Stack after pushing '100' (100 expected): 100

Top (100 expected): 100

Stack after pushing '200' and '300' (100 200 300 expected):

100

200

300

Top (300 expected): 300

Stack after pop (100 200 expected):

100

200

Top (200 expected): 200

Peeked value (200 expected): 200

Top (200 expected): 200

Testing if the stack is empty...

The stack is not empty. Size (2 expected): 2

Stack after pop (100 expected): 100

Top (100 expected): 100

Popping the last value from the stack.

Testing if the stack is empty...

The stack is empty. Size (0 expected): 0

Testing with random stack of positive and negative numbers, including large values...

Stack after pushing 1 random number: -68829

Top: -68829

Stack after pushing a random number (60) of random numbers:

-68829
87187
36751
-14136
-37058
67964
-65382
21012
-91327
98635
-58774
14515
-67308
4697
39622
-41859
81316
-94111
51478
83238
51013
9589
39053
-15022
-35119
-35700
-56015
76869
40810
-40611
36767

-68855
-10799
-57436
16022
-14087
76267
-97299
71861
-18993
-93007
-45190
99482
-57203
92754
60443
62364
51938
-61955
-70366
76038
34441
-44545
67494
71540
35985
24383
52469
-77887
-47289
88270

Top: 88270

Stack after popping a random number (25) of times:

-68829
87187
36751
-14136
-37058
67964
-65382
21012
-91327
98635
-58774
14515
-67308
4697
39622
-41859
81316
-94111
51478
83238
51013
9589
39053
-15022
-35119
-35700
-56015
76869
40810
-40611
36767
-68855

-10799
-57436
16022
-14087

Top: -14087

Peeked value: -14087

Top: -14087

Testing if the stack is empty...

The stack is not empty. Size (expected 36): 36

End of testing.

Part 2

Stack after pushing 5 names:

Nancy
Madeline
Daniel
Charles
Andy

Size: 5

Peek at top of stack: Nancy

Stack after pushing William and Yasmin:

Yasmin
William
Nancy
Madeline
Daniel

Size: 5

Peek at top of stack: Yasmin

Checking to make sure the bottom of the stack dropped out:

Current item: Yasmin at element 0	Next address: LinearNode@2a487f4b
Current item: William at element 1	Next address: LinearNode@7d2c0166
Current item: Nancy at element 2	Next address: LinearNode@3b915c3f
Current item: Madeline at element 3	Next address: LinearNode@29181309
Current item: Daniel at element 4	Next address: null

Part 3

Enqueued to front: 100

Contents:

100

Size: 1 Front: 100 Rear: 100

Enqueued to rear: 200

Contents:

100

200

Size: 2 Front: 100 Rear: 200

Enqueued to front: 300

Contents:

300

100

200

Size: 3 Front: 300 Rear: 200

Enqueued to rear: 400

Contents:

300

100

200

400

Size: 4 Front: 300 Rear: 400

Enqueued to front: 500

Contents:

500

300

100

200

400

Size: 5 Front: 500 Rear: 400

Enqueued to rear: 600

Contents:

500

300

100

200

400

600

Size: 6 Front: 500 Rear: 600

Dequeued from front: 500

Contents:

300

100

200

400

600

Size: 5 Front: 300 Rear: 600

Dequeued from rear: 600
Contents:
300
100
200
400
Size: 4 Front: 300 Rear: 400

Dequeued from front: 300
Contents:
100
200
400
Size: 3 Front: 100 Rear: 400

Dequeued from rear: 400
Contents:
100
200
Size: 2 Front: 100 Rear: 200

Dequeued from front: 100
Contents:
200
Size: 1 Front: 200 Rear: 200

Dequeued from rear: 200
The deque is now empty.
Size: 0
